SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING - CSA17 ARTIFICIAL INTELLIGENCE LAB

Day 1 - Program 8	Course Outcome: CO4	Language: Python 3

Question

Write the python program to implement DFS.

Aim

To implement the Depth First Search graph traversal.

Algorithm / Explanation

DFS goes as deep as possible along a branch before backtracking. It is implemented recursively (call stack) and iteratively (explicit stack).

Time complexity O(V + E).

Program

```
# Depth First Search (DFS) - recursive and iterative
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

def dfs_recursive(graph, node, visited=None):
    if visited is None:
        visited = []
    visited.append(node)
    print(node, end=" ")
    for neighbour in graph[node]:
        if neighbour not in visited:
            dfs_recursive(graph, neighbour, visited)
    return visited

def dfs_iterative(graph, start):
    visited, stack = [], [start]
    while stack:
        node = stack.pop()
        if node not in visited:
            visited.append(node)
            stack.extend(reversed(graph[node]))
    return visited

print("DFS (recursive) from A:")
dfs_recursive(graph, 'A')
print("\nDFS (iterative) from A:", " ".join(dfs_iterative(graph, 'A')))
```

Output

```
DFS (recursive) from A:
A B D E F C
DFS (iterative) from A: A B D E F C
```

Result

Thus the Python program was executed successfully and the output was verified.